The 21st INFORMS Workshop on Data Mining and Decision Analytics

October 31, 2026, San Francisco, CA
Moscone Center South, 747 Howard Street
Level 2 | Main Room 214–215 (combined) | Breakouts 211, 212, 213

Organized by the INFORMS Data Mining Society (DMS)

Workshop Co-Chairs
Hairong Wang, The University of Texas at Austin
Hadis Anahideh, University of Illinois Chicago
Jiachang Liu, The Pennsylvania State University
Adam Meyers, University of Miami

Workshop Schedule At-A-Glance

All times are San Francisco local time (PDT).

Time	Room(s)	Event
07:30–08:00	214–215	Workshop registration and welcome
08:00–09:15	211, 212, 213, 214–215	Parallel Sessions A1–A4
09:15–09:30	—	Break
09:30–10:30	214–215	Keynote 1: Timothy Chan, University of Toronto
10:30–10:45	—	Break
10:45–12:00	214–215	Best Paper Competition: Theoretical Track
10:45–12:00	211, 212, 213	Parallel Sessions B1–B3
12:00–13:00	214–215	Lunch
13:00–14:15	214–215	Best Paper Competition: Applied Track
13:00–14:15	211, 212, 213	Parallel Sessions C1–C3
14:15–14:30	—	Break
14:30–15:30	214–215	Keynote 2: Weijie Su, University of Pennsylvania
15:30–15:45	—	Break
15:45–17:00	214–215	INFORMS DMS Data Challenge
15:45–17:00	211, 212, 213	Parallel Sessions D1–D3
17:00	214–215	Workshop Conclusions and Closing Remarks

Keynote 1 09:30–10:30 · Main Room 214–215

Conformal inverse optimization

Timothy Chan
University of Toronto

Abstract

Inverse optimization is increasingly used to estimate unknown parameters in an optimization model based on decision data. However, when such ”point estimates” are used to prescribe downstream decisions, the resulting decisions may be of low-quality and misaligned with human intuition, and thus less likely to be adopted. To tackle this challenge, we propose a novel decision recommendation pipeline that learns an uncertainty set for the unknown parameters and then solves a robust optimization model to prescribe new decisions. We show that the suggested decisions can achieve bounded optimality gaps, as evaluated using both the ground-truth parameters and human perceptions. Our method demonstrates strong empirical performance compared to the standard inverse optimization pipeline. Finally, we perform a case study where we apply this new pipeline to provide delivery route recommendations in Toronto, Canada. Our approach achieves a significantly higher delivery path adherence rate than current industry practices without compromising service quality. Moreover, our method provides a better trade-off between absolute and perceived decision quality than baselines under various realistic scenarios, including cases with model mis-specification and data scarcity.

Biography

Timothy Chan is the Associate Vice-President and Vice-Provost, Strategic Initiatives, and a Professor in the department of Mechanical and Industrial Engineering at the University of Toronto. His primary research interests are in operations research, optimization, and applied machine learning, with applications in healthcare, medicine, sustainability, and sports. He holds editorial roles in several journals including Operations Research, Management Science, and M&SOM. Recent honours include the 2026 JJ Berry Smith Doctoral Supervision Award from the University of Toronto, the 2026 CORS Award of Merit, the 2026 Innovative Application in Analytics Award from the INFORMS Analytics Society, the 2025 CORS Practice Prize, and finalist for the 2025 INFORMS Wagner Prize. Professor Chan received his B.Sc. in Applied Mathematics from UBC and his Ph.D. in Operations Research from MIT. Before coming to Toronto, he was an Associate in the Chicago office of McKinsey and Company.

Keynote 2 14:30–15:30 · Main Room 214–215

Alignment in Large Language Models: Statistical and Game-Theoretic Perspectives

Weijie Su
University of Pennsylvania

Abstract

Large language models (LLMs) are predominantly aligned with human preferences through reinforcement learning from human feedback (RLHF). In this talk, we explore the theoretical foundations of LLM alignment through the intertwined lenses of statistics and game theory. First, we show how the current formulation of RLHF induces a systematic bias we call preference collapse, and how this can be mitigated by introducing a tailored regularization term into the reward function. Next, we expose a fundamental bottleneck of reward-based alignment, demonstrating that cyclic human preferences cannot be faithfully represented by scalar reward models such as the Bradley-Terry model. More precisely, we establish that such cyclic inconsistencies give rise to a lower bound on the approximation error of any scalar reward fitting. Shifting to a game-theoretic perspective, we focus on Nash learning from human feedback and establish several social choice desiderata for this approach to alignment, including the preservation of preference diversity through the emergence of mixed strategies. Finally, we show that the zero-sum game approach generally cannot perfectly match a target preference distribution as a unique Nash equilibrium.

Biography

Weijie Su is a Professor of Statistics and Data Science at the Wharton School of the University of Pennsylvania, where he co-directs Penn Research in Machine Learning. His research examines the mathematical and statistical foundations of artificial intelligence, including large language models, privacy-preserving data analysis, high-dimensional inference, and optimization. He earned his Ph.D. in Statistics at Stanford University in 2016 and his bachelor's degree in Mathematics at Peking University in 2011. His honors include the 2026 COPSS Presidents' Award, the 2022 SIAM Early Career Prize in Data Science, the 2022 IMS Peter Gavin Hall Early Career Prize, a Sloan Research Fellowship, and an NSF CAREER Award. He is a Fellow of the Institute of Mathematical Statistics and a founding co-editor of Statistical Learning and Data Science.

A1 · Conformal Inference and Distributional Risk

08:00–09:15 · Room 211

Session Chair: Shuyi Chen (Carnegie Mellon University)

Beyond Predicting Responses: Conformal Inference for Latent Distributional Parameters

Minxing Zheng, Wenbin Zhou, Shixiang Zhu
Carnegie Mellon University

From Prescription to Certification: Distribution-Free Risk Assessment for Data-Driven Optimization Decisions

Wenbin Zhou, Agni Orfanoudaki, Shixiang Zhu
Carnegie Mellon University; University of Oxford

Learning Polyhedral Conformal Sets for Contextual Robust Optimization

Shuyi Chen, Wenbin Zhou, Shixiang Zhu
Carnegie Mellon University

A2 · Large Language Models and Human Expertise

08:00–09:15 · Room 214–215

Session Chair: Prakhar Saxena (University of Pennsylvania)

Ranking Large Language Models with Nonparametric Prompts

Zebin Wang, Yi Han, Ethan X. Fang, Lan Wang, Junwei Lu
Harvard University; Columbia University; Duke University; University of Miami

Escaping the Nash Trap: Structural Estimation and Alignment of Strategic Reasoning in Large Language Models

Jiannan Xu, Yongkang Duan, Jane Yi Jiang, Jiding Zhang
University of Maryland; Beijing Normal University; The Ohio State University; Arizona State University

What remains after LLMs: technical knowledge moves from hubs to niches.

Neha Sharma, Simin Li
University of Pennsylvania; Tulane University

Where does human expertise still matter? Evidence from AI and Human Answers on Stack Overflow

Neha Sharma, Prakhar Saxena
The Wharton School, University of Pennsylvania

A3 · Reinforcement Learning and Sequential Search

08:00–09:15 · Room 212

Session Chair: Nipun Galande (Virginia Tech)

Hybrid Learning-Optimization for Multi-Agent Bayesian Search in Naval Operations

Nipun Galande, Anna Svirsko, Daphne Skipper, Esra Buyuktahtakin Toy
Virginia Tech; United States Naval Academy

CausalGDP: Causality-Guided Diffusion Policies for Reinforcement

Xiaofeng Xiao, Xiao Hu, Yang Ye, and Xubo Yue
Northeastern University

Safety-Regularized Multi-Agent Reinforcement Learning for Fuel-Limited Naval Search Against Adaptive Targets

Seung Jin Choi, Elson Cibaku, Daphne Skipper, Anna Svirsko, İ. Esra Büyüktahtakın
Virginia Tech; State University of New York at Plattsburgh; United States Naval Academy

A4 · Statistical Learning and Experimental Decisions

08:00–09:15 · Room 213

Session Chair: Zewei Deng (University of Minnesota)

Causal Discovery from Heteroscedastic Stochastic Dynamical Systems under Imperfect Physical Model
Jianhong Chen, Naichen Shi, Xubo Yue
Northeastern University; Northwestern University

Estimating Influence Functions for Causal Stochastic Optimization
Angela Zhou
University of Southern California

Funnel-Aware Policy Learning for Online Experiments
Haozhe Xu, Kan Xu
The University of Hong Kong; Arizona State University

Distributionally Robust Inference for Delayed Point Processes
Zewei Deng, Shuyi Chen, Shixiang Zhu, Liyan Xie
University of Minnesota; Carnegie Mellon University

Best Paper Competition · Theoretical Track

10:45–12:00 · Main Room 214–215

Decision Analytics for Service-Induced Congestion in Memory-Constrained LLM Serving
Ruicheng Ao, Jing Dong, Gan Luo, David Simchi-Levi
Massachusetts Institute of Technology; Columbia University; Peking University

Sample Complexity for Markov Decision Processes and Stochastic Optimal Control with Static Risk Measures
Cristian Chávez, Yan Li
Pontificia Universidad Católica de Chile; Texas A&M University

Learning to Communicate Under Alignment Uncertainty: A Bayesian Persuasion Bandit Approach
Wenqian Xing, Ramesh Johari
Stanford University

Policy-Aware Design of Large-Scale Factorial Experiments *(Virtual presentation)*
Xin Wen, Xi Chen, Will Wei Sun, Yichen Zhang
New York University; Purdue University

B1 · Clinical Prediction and Patient Heterogeneity

10:45–12:00 · Room 213

Session Chair: Carol Gao (Massachusetts Institute of Technology)

Robustifying and Selecting Cohort-Appropriate Prognostic Models under Distributional Shifts
Dimitris Bertsimas, Carol Gao, Angelos Koulouras, Georgios Antonios Margonis
Massachusetts Institute of Technology; Amazon; Charité - Universitätsmedizin Berlin; Memorial Sloan Kettering Cancer Center

Enhancing Chronic Disease Management by Identifying Patient Comorbidity Heterogeneity through a Novel Probabilistic Model with Variational Inference
Tingting Zhao, Jennifer Dy, James Ross
University of Rhode Island; Northeastern University; Brigham and Women's Hospital

Beyond Mean Dice: Transition-Aware Evaluation for Short-Horizon Tumor Forecasting
Rohit Khanna
The University of Texas at Austin

B2 · Forecasting and Optimization for Complex Systems

10:45–12:00 · Room 211

Session Chair: Chengyi Cai (Carnegie Mellon University)

Multistage Power Grid Capacity Expansion Planning under Uncertain AI Data Center Demands
Chengyi Cai, Wei Gu, Shixiang Zhu, Peter Zhang
Carnegie Mellon University

Multistage Adaptive Robust Optimization for Long-term Energy Infrastructure Resilience Planning
Wei Gu, Shuyi Chen, Shixiang Zhu, Ramteen Sioshansi
Carnegie Mellon University

Enhancing Stock Market Forecasting with Engagement-Aware TikTok Sentiment and Foundation Time-Series Model
Luisa Rosa, Nick Trinh, Yanjun Li
Fordham University

Algorithms for Collaborative Truck-Multi-Drone Delivery in the En-Route Model
Jerry Mathew Oommen, Kelin Luo
University at Buffalo

B3 · Robust and Distributed Machine Learning

10:45–12:00 · Room 212

Session Chair: Ying Lin (University of Houston)

Robust Online Multi-task Learning Under Data Corruption
Tanapol Kosolwattana, Huazheng Wang, Ying Lin
University of Houston; Oregon State University

Learning What Not to Impute: An Uncertainty-Aware Diffusion Framework for Meaningful Missingness
Lixing Zhang, Yidong Ouyang, Weifu Li, Shixiang Zhu, Guang Cheng, Liyan Xie
University of Minnesota; University of California, Los Angeles; Carnegie Mellon University

Decentralized Learning Architectures for Scalable and Privacy-Constrained AI Systems
Samridhi Vats, Pratik Borkar, Sai Mona Duvappu, Abhishek Krovvidi, Saurav S Borah
Purdue University

Best Paper Competition · Applied Track

13:00–14:15 · Main Room 214–215

The Right Amount of Wrong: Error Injection for AI Oversight
Kacey Fang, Greg Powell, Jeffery Painter, Andrew Bate, Michael Lingzhi Li
Harvard Medical School; GSK; London School of Hygiene & Tropical Medicine; Harvard Business School

RegimeUnfold: Forecasting with an Evolving Latent Regime Space
Manrui Jiang, Jingru Huang, Yong Chen, Chen Zhang
Tsinghua University; University of Iowa

Balancing Planning Myopia and Forecast Error: Decision-Focused Sequential Learning
Shuai Wang, Shuyi Chen, Shixiang Zhu
Duke University; Carnegie Mellon University

Causal LLMs-guided Graph Neural Networks for Attribution-Enhanced Conversion Rate Prediction *(Virtual presentation)*
Tao Xue, Yanwu Yang
Huazhong University of Science and Technology

C1 · Industrial Monitoring and Adaptive Optimization

13:00–14:15 · Room 211

Session Chair: Runxiong Wu (University of Wisconsin–Madison)

Diffuse to Detect: Single-Step Predicted-Noise Representations for Reliable and Efficient UAV Anomaly Detection

Mingze Gong, Rui Wang, and Juan Du
The Hong Kong University of Science and Technology (Guangzhou); The Hong Kong University of Science and Technology

Interpretable Process Monitoring via Sequential Latent Decomposition for Multiple Quality Concerns

Runxiong Wu, Andi Wang
University of Wisconsin-Madison

Error-bounded Multistage Bayesian Optimization

Kangan Chen, Siqin Yang, Zhenyu Kong, Hao Yan, Andi Wang
University of Wisconsin-Madison; Arizona State University; Virginia Tech

Integrating Production Flow Dynamics with Maintenance Decision Making through Structured Reinforcement Learning

Han Hu, Shiyu Zhou, Jingshan Li
University of Wisconsin-Madison; City University of Hong Kong

C2 · Predictive Analytics and Sustainability

13:00–14:15 · Room 212

Session Chair: Cristobal Heredia Galleguillos (University of South Florida)

From Wildfire Forecasts to Relief and Recovery Pressure: A Decision-Aware Physics-AI Framework for Wildland-Urban Fires

Yao Zhao, Soohyun Cho, Wayne Wu, Jaeyeon Kim
Southern Illinois University Edwardsville; Rutgers University

ESG-SCAPE: Measuring Category Misalignment between Internal ESG Concerns and Prior Plan Commitments

Xiaochuan Ma, Xinyu Lu
Zhejiang University; The Chinese University of Hong Kong, Shenzhen

CEDAR: CUDA-Enabled Decision-Tree Algorithm for Regression

Cristobal Heredia Galleguillos, Yifan Li, Pedro Chumpitaz Flores, Kaixun Hua
University of South Florida; University of Illinois Urbana-Champaign

C3 · Learning and Language Models for Optimization

13:00–14:15 · Room 213

Session Chair: Tinghan Ye (Georgia Institute of Technology)

Democratizing Large-Scale Re-Optimization with LLM-Guided Model Patches
Tinghan Ye, Arnaud Deza, Ved Mohan, El Mehdi Er Raqabi, Pascal Van Hentenryck
Georgia Institute of Technology

Mathematical Formulation of Attention Architecture for Surrogate Optimization
Minseo Lee, Esra Buyuktahtakın
Virginia Tech

Balancing Solution Quality and Optimization Runtime in Constraint Learning
Jiaxiao Fang, Thiago Serra
University of Iowa

INFORMS DMS Data Challenge

15:45–17:00 · Main Room 214–215

Forecasting Infrastructure Resilience Under Extreme Weather

Team Blue Devil
Shuai Wang, Yuhan Wei, Lang Peng
Duke University
Zhengdong Li
Central University of Finance and Economics

Team SIOL
Amirhossein Eskorouchi, Abhro Shoma Pias, Dongmin Ethan Kang, Munirul Alam
Mississippi State University

Team the Overfitters
Abraham Jesus Bohorquez Gomez, Paula Andrea Torres Prieto, Juan Jose Murillo Aristizabal, Esteban Ladino Nieto
Universidad de los Andes

Team 404 NOT FOUND
Haoran Guo, Lingjie Zhao, Wei Zhao
Tsinghua University

D1 · Social Analytics and Organizational Decisions

15:45–17:00 · Room 212

Session Chair: Miriam Zhou (Carnegie Mellon University)

Who Receives Prompt Help? Ideal Victimhood and Timely Support in Digital Help-Oriented Communities

Yanhan (Savannah) Tang
Southern Methodist University

Measuring the Evolution of Urban Informal Registers During a Societal Disruption Using Social Media Data

Tristan Heitger, Ann-Kathrin Meyer
University of Münster

Mitigating Organizational Bias in Small and Medium-Sized Enterprises: An Artificial Intelligence Framework for Improving Productivity and Accountability

Joy Atagamen, Abiodun Modupe
Houston Christian University; University of Pretoria

Clinical Capacity Requirements for Adopting Medical AI Updates under FDA's Predetermined Change Control Plan

Miriam Zhou
Carnegie Mellon University

D2 · Predictive Modeling and Simulation

15:45–17:00 · Room 213

Session Chair: Mohammadmahdi Ghasemloo (Texas A&M University)

FunSearch and LLM-Guided Discovery of Interpretable Dengue Models for Public Health Decision Support

Sushyanth Sridhar, Michael Robert, Esra Buyuktahtakin
Virginia Tech

MetaboLLM: a metabolomics-specialized large language model for biochemical knowledge integration and predictive metabolite graph construction

Dohyun Ku, Min Gu Kwak, Francisco J. Pasquel, Jing Li
Georgia Institute of Technology; Capital One; Emory University

Quantifying and Attributing Submodel Uncertainty in Stochastic Simulation Models and Digital Twins

Mohammadmahdi Ghasemloo, David J. Eckman, Yaxian Li
Texas A&M University; Intuit Inc.

Monotonic Mixture-of-Experts for Regime-Dependent Monotonic Regression

Yating Fang, Qianqian Zhao, Ryan Sills, Ahmed Aziz Ezzat
Rutgers University

D3 · Data Collection and Adaptive Decision Learning

15:45–17:00 · Room 211

Session Chair: Arian Aflaki (University of Pittsburgh)

Counterfactually Stable Attribute-Based Demand Learning for Dynamic Assortments

Arian Aflaki, Deniz Eskandani, Su-Ming Wu
University of Pittsburgh; Oracle Retail Global Business Unit

Pessimistic Off-Policy Learning under Covariate Shift and Local Differential Privacy

Jongmin Mun, Xiaocong Xu, Yingying Fan
University of Southern California

The Greedy Advantage in Finite-Horizon Bandits

Kai Zhou, Michael Lingzhi Li, Kai Wang
Tsinghua University; Harvard Business School

Bet on Features: Anytime-Valid and Feature-Aware Auditing of Conditional Quantile Forecasters

Ivane Antonov, Sohom Mukherjee, Richard Pibernik, Yo Joong Choe
Julius-Maximilians-Universität Würzburg; Zaragoza Logistics Center; INSEAD